

Protocol: Seven Attributes of Highly Effective Scopes

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It is difficult to find a design that does not include one or more lower-speed serial buses such as JTAG, I²C, SPI, CAN, LIN, FlexRay, RS-232, Mil-std 1553 or a proprietary variant. Furthermore, many products are incorporating higher-speed serial buses such as USB, MIPI and PCIe for data transport.

Protocol apps have become an important scope capability for good reason. With the right scope and protocol application combination, teams can gain quick insight to resolve issues. While all major scope vendors offer protocol decode and triggering applications, the applications vary in capability. So when evaluating an oscilloscope that will include a protocol application, here are seven attributes that should be considered.

1 What protocols are supported and to what degree?

If you have ever hand-decoded waveforms, you will immediately understand the value of real-time decode. Protocol apps are typically offered as software options, and can be added to an existing scope or ordered with a new scope. So when selecting a scope, check the datasheet to see what protocols are supported on a particular scope or a specific scope family.

The datasheet will also typically have good detail on the degree of support offered. Most vendors offer free short-term trial licences. Some vendors have a previously acquired example

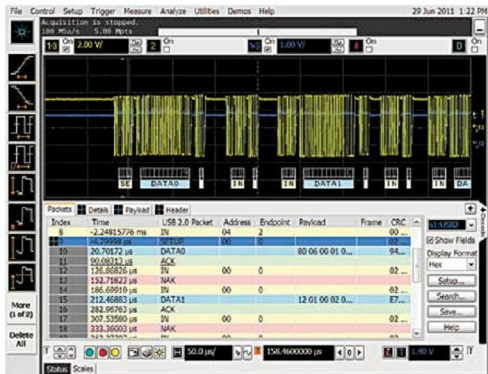


Fig. 1: Protocol lists enable users to quickly move between physical and protocol layers with the advantage of seeing packet detail in the list while zoomed out as shown in this USB decode example

that you can quickly load, or hook up to your own target and try out.

Pull up the application on the scope, to determine how well a particular protocol is supported. For example, if you are using SPI, what is the fastest data rate that is supported? Does the application support 2-, 3- and 4-wire SPI, or just a subset? If you are using USB 2.0, does the application support low-, full- and high-speed versions of the spec as well as HSIC? If using I²C, does the application support PC where the read/write bit is included in the address field?

Often, there is a need to look at simultaneous decode of more than one serial bus. How well does the scope

allow you to set up decode of multiple buses, navigate between buses or change which bus is being used as the trigger source?

2 How easy is it to configure for decode?

Engineers excel at problem solving. Anytime too much brain power or time is required for a task, engineers will find another, less taxing method of attacking a problem. Setting up a scope to take a protocol measurement should be something that you can do in a minute or less.

Configuring the scope for protocol decode involves selecting which channels are probing specific serial signals,